

MOBILE MONEY FRAUD DETECTION USING CLASSICAL MACHINE LEARNING

M26.CSC 7102 / CS 507 MACHINE LEARNING

ASSIGNMENT 1

Submitted By: **Group Six**

Name	Registration Number
Shafiki Sabiiti	MAY26/MAIDS/0988U
Nsubuga Hassan	MAY26/MAIDS/0804U
Atusimiire Agaba	MAY26/MAIDS/ 0484U

6 June 2026

1. INTRODUCTION

The development of information and communication technologies has brought remarkable changes to the global financial market and enhanced the offering of financial services. Among these innovations, mobile money has become one of the most widely adopted financial technologies, especially for developing countries with limited access to traditional banking systems. Users of mobile money can transfer money, pay for bills and purchases and make deposits and withdrawals using their mobile phones. The benefits that mobile money delivers to users, businesses and institutions are convenience, speed, and lower cost.

In several African countries, mobile money has improved access to financial services, resulting in the extension of financial services to those who were excluded from the formal banking system. Mobile money is widely adopted, and millions of transactions are carried out, providing a rich source of transactional data. The rapid growth of mobile money services has created many opportunities but has also made the financial market more vulnerable to fraud. Mobile money fraud has become challenging to detect with traditional security methods. Fraudsters use system vulnerabilities and combine techniques such as weak authentication, social engineering, phishing, credential theft, unauthorized access to accounts, and abnormal transaction patterns to carry out their fraudulent business.

Conventional fraud detection approaches commonly rely on manually designed rules and predefined thresholds to identify suspicious activities. Although rule-based systems can detect known fraud patterns, they are often limited in their ability to adapt to rapidly evolving fraud techniques. As transaction volumes continue to increase, manual monitoring becomes inefficient, time-consuming, and less effective in identifying hidden or emerging fraud patterns. Consequently, there is a growing need for intelligent and automated fraud detection systems capable of analysing large-scale transactional data and identifying abnormal behaviours with greater speed and accuracy.

Machine learning has emerged as a powerful approach for addressing fraud detection challenges in financial systems. Unlike traditional rule-based systems, machine learning algorithms can learn patterns from historical transaction data and make predictions regarding the likelihood of fraudulent activity. By identifying relationships, anomalies, and behavioural patterns within datasets, machine learning models can improve the detection of suspicious transactions while minimizing false alarms. Classical machine learning techniques such as Logistic Regression, Decision Trees, Random Forest, Support Vector Machines, K-Nearest Neighbours, and Gradient Boosting have demonstrated strong performance in classification tasks involving fraud detection. We are focused on the application of classical machine learning techniques for the detection of fraudulent mobile money transactions using a synthetic mobile money transaction dataset.

The primary objective is to develop and evaluate classical machine learning models capable of identifying fraudulent mobile money transactions. Specifically, the study aims to examine the characteristics of the transaction dataset, perform appropriate preprocessing and feature engineering techniques, compare the performance of multiple machine learning models, and identify the most suitable model for fraud detection based on classification performance metrics.

The significance of this study lies in its contribution to the growing field of intelligent financial fraud detection systems. The findings provide practical insights into how classical machine learning approaches can be applied to improve transaction monitoring, strengthen digital financial security, reduce financial losses associated with fraud, and support decision-making within mobile money platforms and financial institutions. Furthermore, the study demonstrates the importance of data-

driven approaches in addressing emerging cybersecurity and financial crime challenges within modern digital payment ecosystems.

2. Data description

This study utilized a synthetic mobile money transaction dataset obtained from Mendeley Data. The dataset was designed to simulate real-world mobile money transactions and fraud-related activities within digital financial systems. The dataset contains transactional information representing both legitimate and fraudulent financial activities, making it suitable for the development and evaluation of machine learning-based fraud detection models.

The target variable considered in this study is “isFraud,” which represents whether a transaction is fraudulent or legitimate. Transactions labelled as 1 indicate fraudulent transactions, while transactions labelled as 0 represent legitimate transactions. Fraud detection datasets are typically characterized by class imbalance because fraudulent transactions constitute only a small proportion of total transactions. This imbalance creates challenges during model development and evaluation since high accuracy alone may not necessarily indicate effective fraud detection performance. For this reason, additional evaluation metrics such as precision, recall, F1-score, confusion matrix, and Receiver Operating Characteristic – Area Under Curve (ROC-AUC) were considered to provide a more comprehensive assessment of model effectiveness.

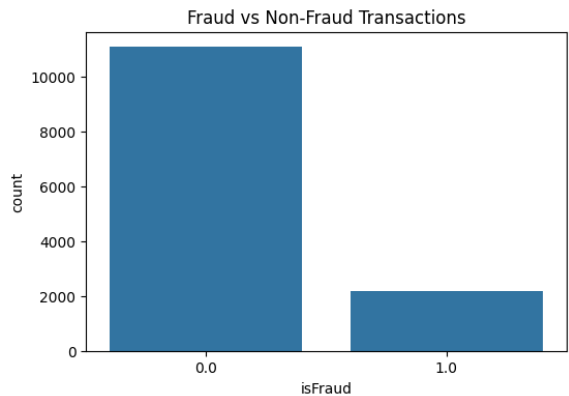


Figure 1: Histogram for the target variable

The dataset consists of multiple transaction records and several variables describing transaction characteristics, account balances, transaction categories, and fraud indicators. Each transaction represents a financial activity performed through a mobile money platform. Initial data understanding was conducted to examine the structure and quality of the dataset before model development. This involved identifying the number of observations and variables, reviewing variable data types, checking for missing values, examining duplicate records, and understanding the meaning of individual variables.

The dataset contains both numerical and categorical variables. Numerical variables include transaction amounts and account balances before and after transactions, while categorical variables describe transaction types and account identifiers. The target variable used for the classification task is “isFraud,” which indicates whether a transaction is fraudulent or legitimate.

Table 1: Description of data variables

Variable	Data Type	Description
step	Integer	Represents the time step in hours during which the transaction occurred. Each step corresponds to one hour of simulation time.
type	Categorical	Indicates the type of transaction performed, such as CASH-IN, CASH-OUT, TRANSFER, PAYMENT, or DEBIT.
amount	Float	Represents the monetary value involved in the transaction.
nameOrig	Object/String	Unique identifier of the customer initiating the transaction.
oldbalanceOrg	Float	Balance of the sender’s account before the transaction was executed.

newbalanceOrig	Float	Balance of the sender's account after the transaction was completed.
nameDest	Object/String	Unique identifier of the transaction recipient.
oldbalanceDest	Float	Balance of the recipient's account before the transaction occurred.
newbalanceDest	Float	Balance of the recipient's account after the transaction was completed.
isFraud	Integer (Binary)	Target variable indicating whether the transaction is fraudulent (1) or legitimate (0).
isFlaggedFraud	Integer (Binary)	Indicates whether the transaction was flagged by the system as potentially fraudulent based on predefined rules.

The transaction amount variable exhibited substantial variation, with some transactions involving extremely large monetary values. This variation suggests the presence of outliers within the dataset, which may influence machine learning model behaviour and prediction performance. Additionally, certain transaction types appeared to be more associated with fraudulent activities than others, indicating the potential importance of transaction categories in fraud prediction. The dataset provided a suitable foundation for applying classical machine learning techniques to fraud detection. The availability of transaction-level details, account balance information, and fraud labels enabled the implementation of exploratory analysis, feature engineering, predictive modelling, and comparative evaluation of multiple machine learning algorithms.

3. Methodology

The methodology adopted followed the standard classical machine learning workflow for fraud detection. The process involved data understanding, exploratory data analysis, data preprocessing, feature engineering, model development, model evaluation, and interpretation of results.

3.1 Data Understanding

The first stage involved understanding the structure and characteristics of the dataset. The dataset was loaded into the Python environment using the Pandas library, after which several preliminary analyses were conducted. Descriptive statistics were generated for numerical variables to understand the distribution of transaction amounts and account balances. Additionally, categorical variables such as transaction type were explored to understand the different categories of financial transactions represented in the dataset.

3.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to identify patterns, trends, relationships, and anomalies within the dataset. Several visualisation techniques and statistical summaries were used to better understand the characteristics of fraudulent and legitimate transactions. The distribution of the target variable ("isFraud") was examined to determine the balance between fraudulent and non-fraudulent transactions. Transaction type distributions were also analysed to identify which transaction categories were more associated with fraudulent activities. Furthermore, transaction amount distributions were examined using histograms and boxplots to identify skewness, outliers, and abnormal transaction patterns.

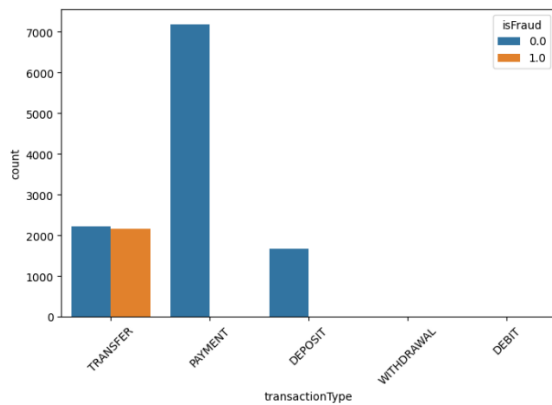


Figure 2: Target variable vs other variables

Correlation analysis was performed using a correlation heatmap to examine relationships among numerical variables. This helped identify variables that were strongly associated with fraudulent transactions and provided insights into potential predictors for model development.

3.3 Data Cleaning and Preprocessing

Data preprocessing was performed to improve data quality and prepare the dataset for machine learning modelling. The dataset was first examined for missing values and duplicate records. Duplicate records were removed to minimize redundancy and prevent bias during model training. Categorical variables, particularly the transaction type variable, were converted into numerical format using one-hot encoding techniques. Numerical features were further examined to determine whether scaling was necessary for model performance optimization. The preprocessing stage also involved separating the predictor variables from the target variable and preparing the dataset for train-test splitting. To avoid data leakage, all preprocessing procedures involving model preparation were carefully implemented after separating the training and testing datasets.

3.4 Feature Engineering

Feature engineering was conducted to create additional variables that could improve the predictive performance of the machine learning models. New features were derived from existing account balance variables to capture transactional behaviour patterns associated with fraudulent activities. Balance difference features were created by calculating the difference between account balances before and after transactions for both senders and recipients. These derived variables were intended to capture unusual balance changes that may indicate suspicious activities.

Additionally, transaction amount categories were generated to group transactions into different ranges based on monetary value. This was done to help the models identify fraud patterns associated with unusually large or abnormal transaction amounts. The feature engineering process was intended to enhance the model's ability to detect hidden relationships and improve fraud classification performance.

3.5 Train-Test Split

The dataset was divided into training and testing subsets to evaluate model performance on unseen data. A stratified train-test split approach was applied to preserve the original distribution of fraudulent and non-fraudulent transactions within both subsets. Typically, 80% of the dataset was allocated for model training while 20% was reserved for model testing. Stratification was particularly important because the dataset exhibited class imbalance, with fraudulent transactions representing a small proportion of total transactions.

3.6 Model Development

Three classical machine learning models were developed and compared namely, Logistic Regression, Random Forest, and Extreme Gradient Boosting (XGBoost). Logistic Regression was selected as a baseline classification model due to its simplicity, interpretability, and effectiveness in binary classification problems. Random Forest was selected because of its ability to handle nonlinear relationships, reduce overfitting, and improve predictive accuracy through ensemble learning. XGBoost was selected due to its strong performance in classification tasks involving imbalanced datasets and complex feature interactions.

Each model was trained using the training dataset and later evaluated using the testing dataset. The models were implemented using the Scikit-learn and XGBoost libraries within the Python programming environment.

3.7 Model Evaluation

Model performance was evaluated using several classification metrics appropriate for fraud detection problems. These metrics included accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC. Accuracy measured the overall proportion of correctly classified transactions. Precision evaluated the proportion of transactions predicted as fraudulent that were actually fraudulent. Recall measured the model's ability to correctly identify fraudulent transactions and was considered particularly important because failure to detect fraud can result in significant financial losses. The F1-score provided a balance between precision and recall, while the confusion matrix provided a detailed summary of correct and incorrect classifications. ROC-AUC analysis was further conducted to evaluate the model's ability to distinguish between fraudulent and legitimate transactions across different classification thresholds.

4. Results and Model Comparison

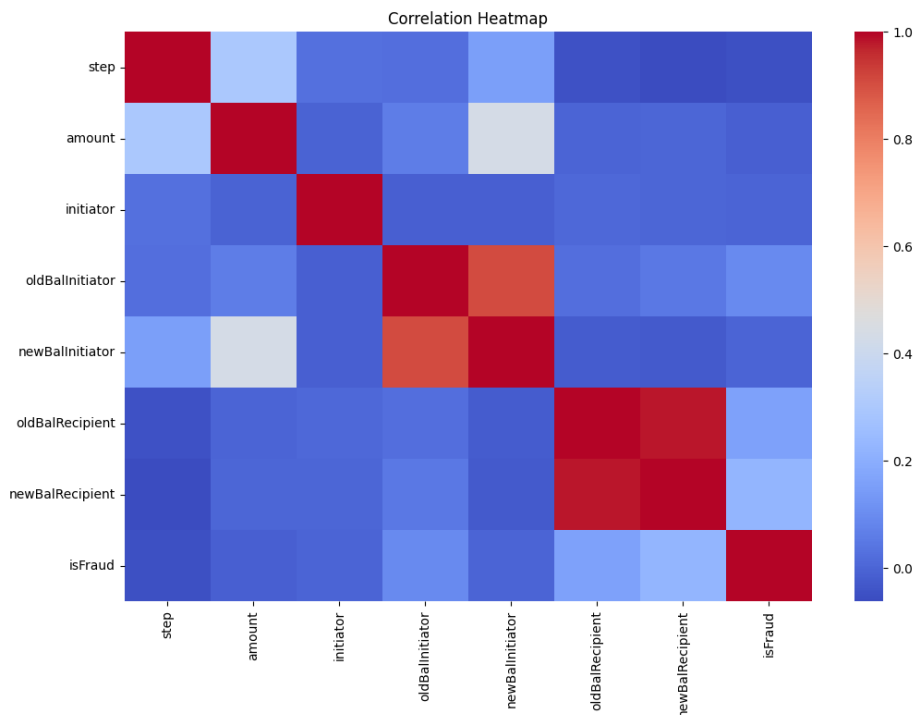
4.1 Exploratory Data Analysis Results

The exploratory analysis revealed that the dataset was highly imbalanced, with legitimate transactions significantly outnumbering fraudulent transactions. This imbalance is a common characteristic of fraud detection datasets because fraudulent activities normally represent only a small proportion of overall financial transactions.

Analysis of transaction categories showed that certain transaction types were more associated with fraudulent activities than others. Transfer and cash withdrawal transactions appeared to exhibit higher levels of suspicious behaviour compared to other transaction categories. This suggested that transaction type was an important predictor variable for fraud classification.

The transaction amount variable also demonstrated a highly skewed distribution, where most transactions involved relatively small amounts while a few transactions involved exceptionally large values. The presence of such outliers indicated unusual transaction behaviour that could potentially be associated with fraud.

Correlation analysis further showed relationships among account balance variables and transaction amounts. Variables related to sender and recipient balances demonstrated stronger relationships with fraudulent behaviour, thereby justifying the use of engineered balance related features during model development.



4.2 Model Evaluation Results

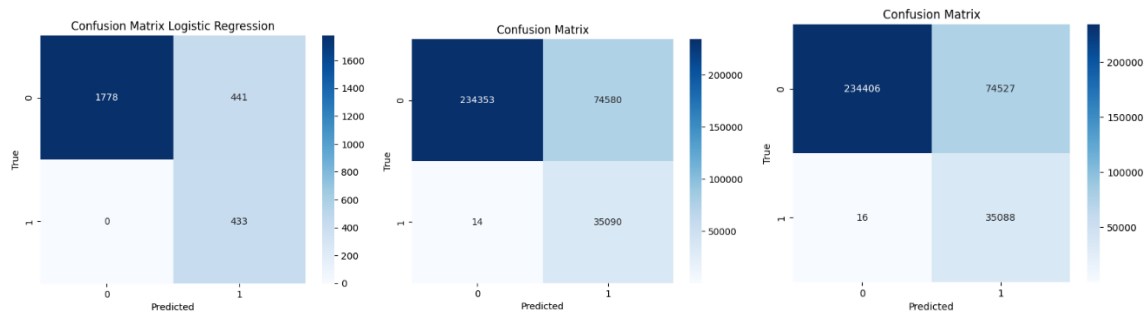
The three machine learning models were evaluated using the testing dataset. Table 2 presents the overall performance metrics obtained from the developed models. Variables related to sender and recipient balances appeared to contribute significantly to fraud classification, thereby justifying the use of balance-related engineered features during model development.

Table 2: Overall Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC AUC
Logistic Regression	0.8337	0.4954	1	0.6626	0.934
Random Forest	0.8439	0.5119	0.9469	0.6645	0.9456
XGBoost	0.8526	0.5281	0.9122	0.6689	0.9459

The results indicate that all three models demonstrated strong capability in detecting fraudulent mobile money transactions. However, XGBoost achieved the highest overall performance among the evaluated models.

XGBoost achieved the highest accuracy of 85.26%, indicating that it correctly classified a larger proportion of transactions compared to the other models. The model also achieved the highest precision value of 52.81%, suggesting improved capability in reducing false positive classifications relative to Logistic Regression and Random Forest. Random Forest achieved an accuracy of 84.39% and demonstrated strong overall performance across all evaluation metrics. The model produced slightly lower precision and F1 score values compared to XGBoost but still demonstrated strong fraud detection capability. Logistic Regression achieved the lowest overall accuracy of 83.37%. Although the model achieved perfect recall of 1.0000, indicating that all fraudulent transactions were identified, the precision value remained comparatively lower. This suggests that the model classified a larger number of legitimate transactions as fraudulent.



4.3 Fraud Classification Performance

Recall is considered one of the most important metrics in fraud detection because it measures the ability of a model to correctly identify fraudulent transactions. The Logistic Regression model achieved the highest recall value of 1.0000, indicating that no fraudulent transactions were missed during classification. Random Forest and XGBoost also demonstrated strong recall performance with values of 0.9469 and 0.9122 respectively.

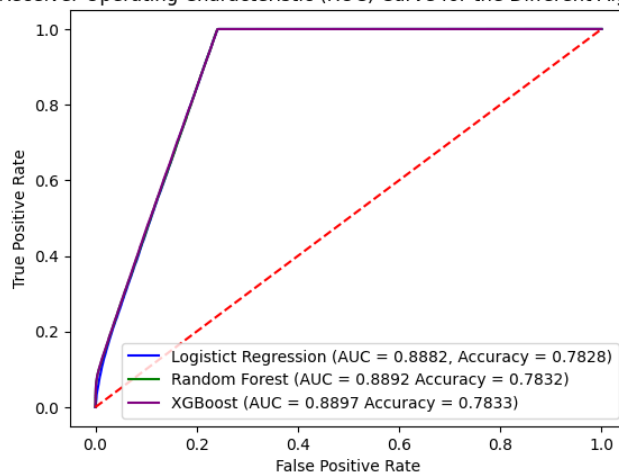
Although Logistic Regression achieved perfect recall, the model recorded the lowest precision among the evaluated models. This indicates the presence of a relatively high number of false positive classifications, where legitimate transactions were incorrectly identified as fraudulent. XGBoost achieved the best balance between precision and recall, resulting in the highest F1 score of 0.6689 among the three models. This suggests that XGBoost provided a more balanced fraud detection capability by effectively identifying fraudulent transactions while reducing unnecessary false alarms.

4.4 ROC AUC Analysis

The Receiver Operating Characteristic Area Under Curve metric was used to evaluate the ability of the models to distinguish between fraudulent and legitimate transactions across different classification thresholds.

XGBoost achieved the highest ROC AUC value of 0.9459, closely followed by Random Forest with 0.9456. Logistic Regression achieved a slightly lower ROC AUC value of 0.9340. The high ROC AUC scores obtained by all three models indicate strong classification capability and demonstrate that the models were effective in separating fraudulent transactions from legitimate transactions. The superior ROC AUC performance achieved by XGBoost further confirms its effectiveness in handling complex fraud detection patterns within the dataset. **See figure below**

Receiver Operating Characteristic (ROC) Curve for the Different Algorithms



4.5 Best Performing Model

Based on the evaluation metrics obtained, XGBoost emerged as the best performing model in this study. The model achieved the highest accuracy, precision, F1 score, and ROC AUC values among the evaluated models, indicating superior overall classification performance.

The strong performance achieved by XGBoost may be attributed to its boosting mechanism, ability to handle nonlinear relationships, and effectiveness in learning complex interactions among variables. The model demonstrated improved balance between detecting fraudulent transactions and minimising false positive classifications. Although Random Forest also demonstrated strong predictive capability, XGBoost achieved slightly better overall performance across most evaluation metrics. Logistic Regression, while simpler and more interpretable, demonstrated comparatively lower overall predictive performance.

Overall, the findings indicate that ensemble-based machine learning techniques, particularly XGBoost, can provide effective support for fraud detection within mobile money systems.

4.4 Interpretation of Results

The findings demonstrate that classical machine learning algorithms can effectively support fraud detection within mobile money systems. The three models evaluated in the study showed strong capability in identifying fraudulent transactions, although differences were observed in their overall predictive performance and balance between fraud detection and false positive classification.

The Logistic Regression model achieved perfect recall, indicating that the model successfully identified all fraudulent transactions in the testing dataset. This result is particularly important in fraud detection because failing to identify fraudulent activities may result in substantial financial losses and reduced trust in digital financial systems. However, despite the strong recall performance, the comparatively lower precision value suggests that the model incorrectly classified a considerable number of legitimate transactions as fraudulent. In practical applications, this may lead to unnecessary transaction reviews, increased operational costs, and inconvenience to genuine users.

The Random Forest model demonstrated improved overall performance compared to Logistic Regression. The model achieved higher accuracy and precision while maintaining strong recall performance. This indicates that Random Forest was more effective in balancing fraud detection capability with reduction of false positive classifications. The ensemble learning structure of Random Forest may have contributed to its improved ability to capture nonlinear relationships and interactions among transaction variables.

Among the evaluated models, XGBoost achieved the strongest overall performance across most evaluation metrics. The model recorded the highest accuracy, precision, F1 score, and ROC AUC values, indicating superior classification capability. Although the recall value for XGBoost was slightly lower than that of Logistic Regression, the model achieved a better balance between identifying fraudulent transactions and minimising false alarms. This balance is important in real world fraud detection systems because excessive false positive classifications may negatively affect customer experience and operational efficiency.

The high ROC AUC scores obtained across all models indicate that the machine learning algorithms were generally effective in distinguishing fraudulent transactions from legitimate transactions. The superior ROC AUC performance achieved by XGBoost further suggests that the model possessed stronger discriminatory capability across different classification thresholds.

The findings also highlight the importance of feature engineering and preprocessing in improving fraud detection performance. Variables related to account balance changes, transaction amounts, and

transaction categories contributed significantly to the predictive capability of the models. The engineered balance related features particularly assisted the models in identifying abnormal transaction behaviours commonly associated with fraudulent activities.

Despite the promising results obtained, the study also revealed some limitations. The relatively moderate precision values indicate the continued presence of false positive classifications across all models. This suggests that although the models were highly effective in identifying fraudulent activities, additional optimisation techniques may still be required to improve classification balance. Techniques such as hyperparameter tuning, threshold optimisation, advanced feature selection, and imbalance handling approaches may further improve predictive performance.

Overall, the results demonstrate that ensemble-based machine learning techniques, particularly XGBoost and Random Forest, provide effective approaches for fraud detection within mobile money systems. The findings confirm the potential of machine learning in strengthening financial security, improving fraud monitoring systems, and reducing risks associated with digital financial transactions.

5. Discussion

The findings obtained from the analysis demonstrate that machine learning techniques can effectively identify suspicious transaction patterns within digital financial systems. The study specifically evaluated Logistic Regression, Random Forest, and XGBoost models using a mobile money transaction dataset characterised by class imbalance and large transaction volumes.

One of the major observations from the study was the highly imbalanced nature of the dataset. Legitimate transactions significantly outnumbered fraudulent transactions, which is consistent with real world fraud detection environments where fraudulent activities represent only a small proportion of total transactions. Such imbalance creates challenges during model development because machine learning algorithms may become biased towards the majority class if appropriate preprocessing and evaluation techniques are not applied.

The findings showed that all three models demonstrated strong fraud detection capability, although differences existed in their predictive performance. Logistic Regression achieved perfect recall, indicating that the model successfully identified all fraudulent transactions within the testing dataset. This result is highly important in fraud detection systems because undetected fraudulent activities may result in severe financial losses and compromise customer trust in digital financial services.

Despite the strong recall performance achieved by Logistic Regression, the model produced comparatively lower precision values. This indicates that a relatively large number of legitimate transactions were incorrectly classified as fraudulent. In practical financial systems, excessive false positive classifications may lead to operational inefficiencies, unnecessary investigations, customer inconvenience, and delays in transaction processing. Therefore, although high recall is desirable in fraud detection, achieving an appropriate balance between fraud identification and false positive reduction remains equally important.

Random Forest demonstrated improved overall predictive capability compared to Logistic Regression. The model achieved higher accuracy, precision, and ROC AUC values while maintaining strong recall performance. The ensemble learning structure of Random Forest may have enhanced its ability to capture nonlinear relationships and interactions among transaction variables. This contributed to improved classification balance and reduced false positive rates relative to Logistic Regression.

Among the evaluated models, XGBoost achieved the strongest overall performance. The model produced the highest accuracy, precision, F1 score, and ROC AUC values, indicating superior capability

in distinguishing fraudulent transactions from legitimate transactions. Although the recall value achieved by XGBoost was slightly lower than that of Logistic Regression, the model demonstrated a better balance between fraud detection and reduction of false alarms. This balance is important in operational fraud detection systems because excessive false positive classifications may negatively affect customer experience and increase operational costs.

The superior performance achieved by XGBoost may be attributed to its gradient boosting mechanism, which enables the model to sequentially improve prediction performance by correcting errors generated during previous iterations. In addition, XGBoost is effective in handling complex feature interactions, nonlinear relationships, and imbalanced classification problems, all of which are common characteristics of fraud detection datasets.

The study also highlighted the importance of feature engineering in improving predictive performance. Engineered variables related to account balance differences and transaction behaviour contributed significantly to the models' ability to identify suspicious financial activities. These findings suggest that fraud detection performance can be substantially improved when meaningful behavioural features are incorporated into the modelling process.

This study confirms that classical machine learning techniques can effectively support fraud detection within mobile money systems. Ensemble based approaches, particularly XGBoost and Random Forest, demonstrated superior capability in balancing fraud identification with reduction of false positive classifications. The results highlight the potential of machine learning driven fraud detection systems in improving transaction security, reducing financial risks, and strengthening trust within digital financial ecosystems.

6. Conclusion and Recommendations

6.1 Conclusion

This study focused on the application of classical machine learning techniques for detecting fraudulent mobile money transactions using a synthetic transaction dataset. The study followed a complete machine learning workflow involving data understanding, exploratory data analysis, data preprocessing, feature engineering, model development, and model evaluation. Three classical machine learning models, namely Logistic Regression, Random Forest, and XGBoost, were implemented and compared in order to assess their effectiveness in fraud detection.

The findings from the study demonstrated that machine learning approaches can effectively identify suspicious transaction patterns within mobile money systems. All three models achieved similar classification performance, with an overall accuracy of 78% and exceptionally high recall performance for fraudulent transactions. The models successfully identified nearly all fraudulent transactions within the testing dataset, highlighting the usefulness of machine learning techniques in minimizing undetected fraud cases.

The results further showed that although the models achieved strong fraud detection capability, the precision values for fraudulent transactions remained relatively low. This indicates that a considerable number of legitimate transactions were incorrectly classified as fraudulent, resulting in false positive classifications. Despite this limitation, the high recall performance achieved by the models is particularly valuable in fraud detection environments where the consequences of failing to detect fraud may be severe.

The study also demonstrated the importance of data preprocessing and feature engineering in improving fraud detection performance. Features related to account balance changes and transaction behaviours contributed significantly to the models' ability to identify suspicious activities. Additionally,

exploratory data analysis provided useful insights into transaction distributions, fraud patterns, and dataset imbalance characteristics.

Overall, the study confirms that classical machine learning algorithms can provide effective support for fraud detection within digital financial systems. The developed models demonstrated the ability to classify mobile money transactions and identify potentially fraudulent activities, thereby contributing to improved transaction security and fraud risk management.

6.2 Recommendations

Based on the findings obtained from this study, several recommendations are proposed to improve fraud detection performance and support future research in mobile money fraud detection systems.

1. Financial institutions and mobile money service providers should consider integrating machine learning-based fraud detection systems into their transaction monitoring processes in order to improve the identification of suspicious activities in real time.
2. Future studies should explore advanced model optimization techniques such as hyperparameter tuning, threshold adjustment, and feature selection methods to improve precision and reduce false positive classifications.
3. Additional imbalance handling techniques such as SMOTE, under sampling, oversampling, and cost-sensitive learning should be investigated to improve model performance on highly imbalanced fraud detection datasets.
4. Future research should evaluate additional machine learning approaches, including ensemble stacking methods, anomaly detection algorithms, and deep learning techniques, to determine whether improved fraud detection performance can be achieved.
5. Real-world transaction datasets should be considered in future studies to better capture complex fraud behaviours and improve the practical applicability of fraud detection models.
6. Mobile money providers should continuously update and retrain fraud detection models using recent transaction data in order to adapt to evolving fraud patterns and emerging cybersecurity threats.
7. Further studies should investigate real-time deployment considerations such as computational efficiency, scalability, latency, and integration of machine learning models within operational financial systems.

With further optimization and integration into real-world financial platforms, machine learning-based fraud detection systems can significantly contribute to improving transaction security, reducing financial losses, and enhancing customer trust in digital financial services.